Giovanni Sullutrone

LLM safety researcher · refusal, evaluation and agents

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I like finding the places where language models behave in ways nobody planned for: a harmless request refused because of the documents around it, or an agent that senses a question is wrong but cannot say why. Most of my papers started from a discussion with researchers or users, or from a claim in the literature that I wanted to test. Lately, I have started looking for the causes inside the models themselves, with interpretability methods.

EDUCATION

PhD in Information and Communication Technologies, *University of Modena and Reggio Emilia* Nov 2023 – Oct 2026

UNIMORE, DBGroup, Department of Engineering "Enzo Ferrari". Supervisor: Prof. Sonia Bergamaschi. Three-year programme, November 2023 to October 2026. Defence expected in early 2027, degree not yet awarded.

MSc in Computer Engineering, *University of Salerno* Oct 2021 – Oct 2023

Artificial Intelligence and Intelligent Robotics track. Final mark 110/110 cum laude (Italian scale, maximum 110).

BSc in Computer Engineering, *University of Salerno* Oct 2018 – Oct 2021

Final mark 110/110 (Italian scale, maximum 110).

RESEARCH EXPERIENCE

PhD Candidate (research focus: LLM safety and evaluation), *UNIMORE, Modena, Italy* Nov 2023 – Oct 2026

- Led two first-author studies of refusal and over-refusal, one in retrieval-augmented generation (COVER) and one in Text-to-SQL (Text-to-Refused-SQL). Designed their controlled prompt conditions, such as the task, the system prompt, a chain-of-thought instruction or an authorization statement, and measured how refusal changed. Both studies were part of ITSERR, a research infrastructure project of the Italian Recovery and Resilience Plan (PNRR) funded by NextGenerationEU.
- Built the evaluation pipelines for these two studies and for the ABISS benchmark. They run open models with vLLM on A100 GPU nodes and call the OpenAI and Gemini APIs. Across the three studies, they evaluated 21 models from 3B to 122B parameters, among them Llama-3-70B, GPT-OSS-120B, GPT-4o-mini, Gemini 1.5 Flash and Gemini 2.0 Flash-Lite, on question answering, translation, summarization and SQL generation.
- Built the scoring stages: Llama-Guard-3-8B flags unsafe passages, and an LLM judge (Mistral-Small-2501) labels refusals. Validated the judge against human labels: two co-authors independently labelled 100 sampled answers, and their labels matched the judge's in all 100 cases.
- In ITSERR's Digital Maktaba work package, worked in the team on OCR for historical Arabic-script documents. The team improved and benchmarked OCR models, built a tool for semi-automatic cataloguing and search, and created DM-LP, a new dataset of frontispieces.
- Outside ITSERR, built the data-collection proof of concept for EpiCDIn, from requirements and planning to the graphical interface and first use. EpiCDIn is a regional feasibility project on AI-guided epigenetics for cardiovascular disease and sudden unexplained death (Emilia-Romagna Region, Dec 2025 – Dec 2026, PI Prof. Rossana Cecchi, with the University of Copenhagen).

Visiting Researcher, *DARE Lab, Athena Research Center, Athens, Greece* (host: Dr Georgia Koutrika) Apr 2025 – Jul 2025

- Designed ABISS, a benchmark that evaluates Text-to-SQL agents on ambiguous and unanswerable requests through multi-turn interaction with simulated users. Three 120B-class open models (GPT-OSS-120B, Qwen3.5-122B-A10B, Nemotron-3-Super-120B) generate and validate the questions and act as the simulated user.

Machine Learning Researcher (research grant), *A.I. Tech S.r.l., Fisciano, Italy* 2023

- Defined the user requirements, then designed and implemented a monitoring and prediction system for public transport management.

SELECTED FIRST-AUTHOR PUBLICATIONS

COVER: Context-Driven Over-Refusal Verification in LLMs Findings of ACL 2025

G. Sullutrone, R. A. Vigliermo, S. Bergamaschi, L. Sala paper code

Introduces context-driven over-refusal: a model rejects a benign task because of the supplied documents, not the request. Evaluates 13 models on 8 tasks and 3 system prompts over the same retrieved documents, with five samples per prompt. Translation and summarization were refused more often than question answering (20% and 18% against 11% on average), and some model and task settings refused every request. GPT-4o-mini refused 78% of hadith translations under an ethical system prompt. A chain-of-thought instruction raised Llama-3-8B refusal on question answering from 25% to 43% under a helpful system prompt. Later preprints cite COVER, among them work that turns context-driven refusal into blocking attacks on RAG systems.

Text-to-Refused-SQL: A Comprehensive Evaluation of LLMs Refusal in Text-to-SQL SEBD 2025
G. Sullutrone, L. Sala, L. Trigiante, S. Bergamaschi paper

Defines schema-driven over-refusal and measures it in 7 models, among them GPT-4o-mini and Gemini 2.0 Flash-Lite, on 319 healthcare databases augmented with synthetic personal and sensitive fields, with five samples per prompt. Under a strict ethical system prompt, Llama-2 refused 58% of questions that used no personal field, while GPT-4o-mini, Gemini 2.0 Flash-Lite and Phi-4 stayed below 1% in every setting. Stating that access to the personal data was authorized raised refusal in almost every setting where a model refused at all, often more than doubling it.

ABISS: Evaluating Text-to-SQL Systems Through Agent Interaction arXiv 2026 · under review, EACL 2027
G. Sullutrone, L. Sala, S. Aftar, G. Koutrika, S. Bergamaschi paper

Proposes a taxonomy of 8 categories of ambiguous and unanswerable database questions. A multi-agent pipeline with ten validation stages and majority votes of three open models generates 7,145 questions on BIRD and Spider. Two experts judged the category correct in 92% of ratings on 52 sampled questions. In multi-turn interaction with simulated users, every agent recognized problematic questions more often than it classified them correctly (best model on BIRD: 70.5% against 59.1%). Even with the correct category supplied, execution accuracy on ambiguous questions averaged only 32% on BIRD and 40% on Spider.

PUBLICATIONS

Twelve works, 2024–2026: five international conference papers, one journal article, one book chapter, four papers at Italian national venues (CEUR-WS) and one preprint under review. First author on five, sole author on one. **Bold**: G. Sullutrone. * equal contribution. † corresponding author.

Preprint under review

1. **G. Sullutrone**, L. Sala, S. Aftar, G. Koutrika, S. Bergamaschi. “ABISS: Evaluating Text-to-SQL Systems Through Agent Interaction.” arXiv:2607.23340, July 2026. Under review for EACL 2027 (ACL Rolling Review).

International conference papers

2. L. Sala, **G. Sullutrone**, S. Bergamaschi. “Text-to-SQL with Large Language Models: Challenges Revisited and New Dimensions.” 15th International Conference on Data Science, Technology and Applications (DATA 2026), pp. 621–628. Position paper. doi:10.5220/0015161100004091
3. L. Sala, R. A. Vigliermo, **G. Sullutrone**, S. Bergamaschi. “DM-LP: A Large Arabic Script Languages Dataset for OCR and Cataloguing Research.” New Trends in Theory and Practice of Digital Libraries (TPDL 2025), CCIS, pp. 124–134. Short paper. doi:10.1007/978-3-032-06136-2_12
4. **G. Sullutrone**†, R. A. Vigliermo, S. Bergamaschi, L. Sala. “COVER: Context-Driven Over-Refusal Verification in LLMs.” Findings of the Association for Computational Linguistics: ACL 2025, pp. 24214–24229. doi:10.18653/v1/2025.findings-acl.1243
5. L. Sala, **G. Sullutrone**, S. Bergamaschi, R. A. Vigliermo. “Arabic Text Enhancement with GPT for Digital Libraries.” 24th ACM/IEEE Joint Conference on Digital Libraries (JCDL 2024). Short paper. Second place, Best Short Paper Award. doi:10.1145/3677389.3702509
6. **G. Sullutrone**, R. A. Vigliermo, L. Sala, S. Bergamaschi. “Sensitive Topics Retrieval in Digital Libraries: A Case Study of ḥadīṯ collections.” Linking Theory and Practice of Digital Libraries (TPDL 2024), LNCS, pp. 51–62. doi:10.1007/978-3-031-72440-4_5

Journal article and book chapter

7. R. A. Vigliermo, **G. Sullutrone**, S. Bergamaschi, L. Sala. “The ‘Digital Maktaba LP’: Proposing a Comprehensive Dataset for Arabic Script OCR Title Pages in the Context of Digital Libraries and Religious Archives.” Umanistica Digitale, no. 22, pp. 77–111, 2026. doi:10.60923/issn.2532-8816/22019
8. S. Bergamaschi, F. Ruozzi, D. Ciccarello, R. A. Vigliermo, L. Sala, A. El Ganadi, **G. Sullutrone**, S. Aftar. “The Digital Maktaba Platform: A Tool for Semi-Automatic Document Cataloguing and Searching.” In The Digital Turn in Religious Studies, V&R unipress, 2025, pp. 395–414. doi:10.14220/9783737018944.395

Italian national venues (CEUR-WS)

9. **G. Sullutrone**†, L. Sala*, L. Trigiante*, S. Bergamaschi. “Text-to-Refused-SQL: A Comprehensive Evaluation of LLMs Refusal in Text-to-SQL.” 33rd Italian Symposium on Advanced Database Systems (SEBD 2025), CEUR-WS Vol-4182.
10. R. A. Vigliermo, **G. Sullutrone**, S. Bergamaschi, L. Sala. “Proposing a Comprehensive Dataset for Arabic Script OCR in the context of Digital Libraries and Religious Archives (Extended Abstract).” Italian Research Conference on Digital Libraries (IRCDL 2025), CEUR-WS Vol-3937.
11. L. Sala*†, **G. Sullutrone***, S. Bergamaschi*. “Text-to-SQL with Large Language Models: Exploring the Promise and Pitfalls.” Italian Symposium on Advanced Database Systems (SEBD 2024), CEUR-WS Vol-3741. Discussion paper.
12. **G. Sullutrone**. “Large Language Models integration in Digital Humanities.” SEBD 2024 Doctoral Consortium, CEUR-WS Vol-3741. PhD research proposal.

SOFTWARE AND DATA

COVER, evaluation pipeline for context-driven over-refusal, with vLLM, OpenAI and Gemini back-ends, a Llama Guard stage and an LLM-judge stage 2025
github.com/giosullutrone/COVER

OTHER EXPERIENCE

Software Development Consultant (freelance), *DataRiver* Mar 2026 – Jul 2026

Built proofs of concept for clinical and healthcare data collection, from the requirements and planning to the graphical interface and its first use, with testing and technical documentation.

AWARDS

Second place, Best Short Paper Award, JCDL 2024 Dec 2024

For "Arabic Text Enhancement with GPT for Digital Libraries" (second author).

SKILLS

Research	LLM safety evaluation, refusal and over-refusal, controlled prompt conditions, LLM judges with human validation, safety classifiers (Llama Guard 3), multi-turn agent evaluation with simulated users, benchmark design, retrieval-augmented generation, Text-to-SQL
Engineering	Python, PyTorch, Hugging Face Transformers, vLLM, OpenAI and Gemini APIs, TRL, LoRA fine-tuning, A100 GPU clusters, Docker, SQL, Git